TERMWISE

Deployment Guide

Rebuild Termwise locally, verify it, and sign in with the demo accounts.

Repository	https://github.com/moneytosms/termwise
Live app	https://termwise-s1vj.onrender.com
Project site	https://moneytosms.github.io/termwise/
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You can rebuild everything locally with the steps below, or open the live deployment at https://termwise-s1vj.onrender.com (same demo logins).

1. Prerequisites

1. Node 24 or later (CI uses 24, the Dockerfile uses 26). `package.json` has no `engines` field.
2. pnpm 12.15.0 (`packageManager` in `package.json`).
3. Docker with the compose plugin, and git.

```
node -v                     # v24 or later
npm install -g pnpm@12.15.0
pnpm -v                     # 12.15.0
docker compose version
git --version
```

2. Run locally

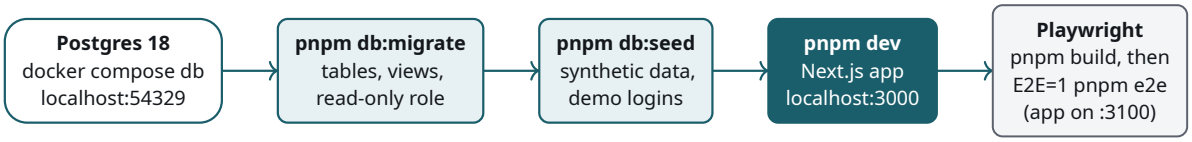


Figure 1. Local setup order: database, migrate, seed, app, then the browser tests.

```
git clone https://github.com/moneytosms/termwise && cd termwise
pnpm install
cp env.example .env.local        # local-only values, no real secrets
set -a; . ./env.example; set +a  # tsx scripts do not read .env.local, so export the same
values
docker compose up -d db          # Postgres 18 on localhost:54329
docker compose ps db             # wait until STATUS shows (healthy)
pnpm db:migrate                  # tables, assistant views, read-only role
termwise_assistant
pnpm db:seed                     # synthetic data and demo logins
pnpm dev                         # then open http://localhost:3000
```

Open http://localhost:3000 and sign in with a login from section 4.

Assistant role: no manual setup. `db:migrate` runs `sql/assistant.sql` (creates the read-only role and views) and sets the role's password from `ASSISTANT_DATABASE_URL` . `env.example` already points at `termwise_assistant` on the same database.

Env var names (values for local runs come from `env.example` ; never commit real secrets):

Name	Required	Purpose
`DATABASE_URL`	yes	App database (set in `env.example`)
`ASSISTANT_DATABASE_URL`	yes	Read-only role the assistant runs as: same database, user `termwise_assistant` . Migration sets the role's password from it
`BETTER_AUTH_SECRET`	yes	Signs sessions and receipt hashes. Changing it breaks verification of issued receipts

Name	Required	Purpose
`BETTER_AUTH_URL`	yes	Public URL of the app as the browser sees it (`http://localhost:3000` locally)
`CRON_SECRET`	for the daily job	Bearer secret for `/api/cron/daily` (blank in `env.example`)
`GOOGLE_GENERATIVE_AI_API_KEY`	optional (one LLM key is enough)	Gemini key (blank in `env.example`)
`GROQ_API_KEY`	optional (one LLM key is enough)	Groq key (blank in `env.example`)
`LLM_PROVIDER`	optional	Preferred provider, `groq` or `google` . If its key is missing, the other keyed provider is used
`LLM_DAILY_CAP`	optional	Max LLM calls per day

Without an LLM key the app runs; only the plain-English assistant and AI-worded risk reasons are unavailable. The assistant says no provider is configured and risk reasons use the rule's own wording.

Health check: `/api/health` returns 200 only when the app database answers and every required var is set, otherwise 503. The body lists each var as a boolean (never values), plus `db` , `assistantDb` , `llm` and `missing` .

Daily job (optional): in GitHub repository settings add secrets `APP_URL` (public URL) and `CRON_SECRET` (the same value as above). The `Daily job` workflow calls `POST /api/cron/daily` at 06:00 IST and can be run by hand from the Actions tab.

Start over with fresh sample data (wipes the volume):

```
docker compose down -v
docker compose up -d db
docker compose ps db              # wait for (healthy)
pnpm db:migrate && pnpm db:seed
```

Or run the whole stack in containers (migrates and seeds on start, app on http://localhost:3000):

```
docker compose up --build
```

3. Verify

Run in the shell where the env was exported (section 2), on a freshly seeded database.

```
pnpm lint
pnpm typecheck                              # tsc --noEmit
pnpm test                                   # vitest run (unit tests)
pnpm build
pnpm exec playwright install chromium       # add --with-deps on a bare Linux machine
E2E=1 pnpm e2e                              # desktop specs, then phone specs, against
pnpm start -p 3100
pnpm tsx scripts/confirm-race.mts           # concurrent admissions get consecutive roll
numbers; a repeat confirm is refused
```

Expected: each command exits 0; vitest and Playwright report all tests passed (counts are in `docs/tdd.md`). The e2e specs and `confirm-race.mts` change seeded data (admissions, attendance), so reset the database (section 2) before re-running or demoing.

Load test (7,800 students reading results at once). Start the server with `E2E=1` so the sign-in rate limiter is off, then in a second terminal:

```
E2E=1 BETTER_AUTH_URL=http://localhost:3000 pnpm start     # terminal 1
BASE=http://localhost:3000 EMAIL=ankit.panda@termwise.demo N=7800 C=200 pnpm tsx scripts/
load-results.mts   # terminal 2
```

`BASE`, `EMAIL`, `N`, `C` are optional script inputs (defaults shown), not app settings.

CI (`.github/workflows/ci.yml`) runs the same steps against Postgres 18 on every push to `main` and on pull requests.

4. Demo logins

Password for every account: `termwise-demo`. All data is synthetic.

Email	Name	Role	Campus
`eng.admin@termwise.demo`	Sunita Rath	Office admin	Engineering College
`cse.faculty@termwise.demo`	Bibhu Prasad Sahoo	Faculty	Engineering College
`cse.faculty2@termwise.demo`	Madhusmita Nayak	Faculty	Engineering College
`me.faculty@termwise.demo`	Ranjan Kumar Jena	Faculty	Engineering College
`subhashree.mohanty@termwise.demo`	Subhashree Mohanty	Student	Engineering College
`prakash.mohanty@termwise.demo`	Prakash Mohanty	Parent	Engineering College
`ankit.panda@termwise.demo`	Ankit Panda	Student	Engineering College
`sarat.panda@termwise.demo`	Sarat Panda	Parent	Engineering College
`pallavi.jena@termwise.demo`	Pallavi Jena	Parent	Engineering College
`mgt.admin@termwise.demo`	Lopamudra Das	Office admin	Management School
`mba.faculty@termwise.demo`	Sandeep Mishra	Faculty	Management School
`farheen.ali@termwise.demo`	Farheen Ali	Student	Management School
`imran.ali@termwise.demo`	Imran Ali	Parent	Management School
`phr.admin@termwise.demo`	Sasmita Behera	Office admin	Pharmacy College
`phr.faculty@termwise.demo`	Debasis Mohapatra	Faculty	Pharmacy College
`phr.faculty2@termwise.demo`	Itishree Swain	Faculty	Pharmacy College
`visiting.faculty@termwise.demo`	Prakash Chandra Rout	Visiting faculty	Engineering + Management
`dual.role@termwise.demo`	Anjali Mohanty	Faculty + office admin	Engineering + Pharmacy
`eng.admissions@termwise.demo`	Smita Pattnaik	Admissions	Engineering College
`eng.exams@termwise.demo`	Nirakar Behera	Exams	Engineering College
`eng.accounts@termwise.demo`	Jyotirmayee Sahu	Accounts	Engineering College
`eng.hr@termwise.demo`	Bikash Chandra Dash	HR	Engineering College
`mgt.admissions@termwise.demo`	Gitanjali Mohapatra	Admissions	Management School

Email	Name	Role	Campus
`mgt.exams@termwise.demo`	Rabindra Nath Satpathy	Exams	Management School
`mgt.accounts@termwise.demo`	Manasi Routray	Accounts	Management School
`mgt.hr@termwise.demo`	Tapan Kumar Swain	HR	Management School
`phr.admissions@termwise.demo`	Pratyusha Nanda	Admissions	Pharmacy College
`phr.exams@termwise.demo`	Santosh Kumar Barik	Exams	Pharmacy College
`phr.accounts@termwise.demo`	Sumitra Pradhan	Accounts	Pharmacy College
`phr.hr@termwise.demo`	Alok Ranjan Mahapatra	HR	Pharmacy College
`cse.hod@termwise.demo`	Dr. Soumya Ranjan Panigrahi	Head of department	Engineering College
`me.hod@termwise.demo`	Dr. Bhabani Shankar Nayak	Head of department	Engineering College
`mba.hod@termwise.demo`	Dr. Smita Rani Pattnaik	Head of department	Management School
`phr.hod@termwise.demo`	Dr. Ashok Kumar Sahu	Head of department	Pharmacy College
`trust@termwise.demo`	Bijay Mishra	Trust	All campuses

Every student and guardian has a login too, `firstname.lastname@termwise.demo` (a number is added when a name is taken), for example `ankit.panda@termwise.demo` for the student and `sarat.panda@termwise.demo` for the parent. The guardian `pallavi.jena@termwise.demo` (parent of Amrita Jena, ENG-24-CSE-0001) receives messages in Odia: sign in and open Messages to read them.

What to try first

- **Office admin:** `eng.admin@termwise.demo` confirms an applicant from **Applicants**, and advances a batch from College setup.
- **Faculty:** `cse.faculty@termwise.demo` marks attendance offline on a phone.
- **Head of department:** `cse.hod@termwise.demo` lists the department's students and approves a CSE leave request.
- **Admissions, exams, accounts and HR:** `eng.admissions@termwise.demo`, `eng.exams@termwise.demo`, `eng.accounts@termwise.demo` or `eng.hr@termwise.demo` each see only their narrow menu; `eng.accounts@termwise.demo` awards a scholarship on the fees page.
- **Visiting and dual-role:** `visiting.faculty@termwise.demo` and `dual.role@termwise.demo` switch the active campus in the sidebar.
- **Trust:** `trust@termwise.demo` compares campuses on the overview and opens the NAAC page.
- **Student and parent:** `ankit.panda@termwise.demo` or `sarat.panda@termwise.demo` opens fees and receipts on the portal.

Try first: sign in as `cse.faculty@termwise.demo` on a phone, open a class from Home, turn on airplane mode, mark it, and turn the network back on. Then sign in as `eng.admin@termwise.demo` to see the sheet, lock marks, record a fee payment and open the at-risk list.

Admissions and staff leave:

- **Confirm an admission (admin).** Sign in as `eng.admin@termwise.demo` and open **Applicants**. The seeded applicant **Ritika Nayak** has every required document verified, so she is ready to confirm. Open her panel and choose **Confirm admission**. The student gets the next roll number in the batch, two fee instalments and a guardian login with the demo password. The other seeded applicants show what is missing, and one applicant can be rejected with a reason.

- **Import applicants (admin).** On **Applicants**, choose **Import CSV**, pick the batch and a file (the template downloads from the dialog). Any bad line stops the whole file, and the dialog lists each line's error.
- **Staff leave (faculty).** Sign in as `cse.faculty@termwise.demo` and open **Leave**. Apply for leave, and see the balances and history. Withdraw the request while it is pending.
- **Approve staff leave (admin).** Sign in as `eng.admin@termwise.demo` and open **Leave**. Approve or reject the pending request from the Engineering campus, with a note. Nobody can decide their own leave.

To try the admission again, reset the database as above.

5. Troubleshooting

1. `pnpm db:migrate` fails with `ECONNREFUSED` or an undefined `DATABASE_URL`: the db is not up or the env is not exported. Run `docker compose ps db` (wait for healthy), then `set -a; . ./env.example; set +a`.
2. Port 54329 or 3000 already in use: stop the other process, or `docker compose down` and retry. Playwright uses port 3100.
3. Sign-in rejected or redirected to the wrong host: `BETTER_AUTH_URL` must equal the URL in the browser (`http://localhost:3000` locally).
4. Playwright or the load test hits rate limits or stale data: run with `E2E=1` (rate limiter off) and reset the database first (section 2).
5. `pnpm exec playwright install chromium` fails on missing libraries: rerun with `--with-deps`.
6. Assistant says no provider is configured: expected without an LLM key. Set `GOOGLE_GENERATIVE_AI_API_KEY` or `GROQ_API_KEY` and restart.